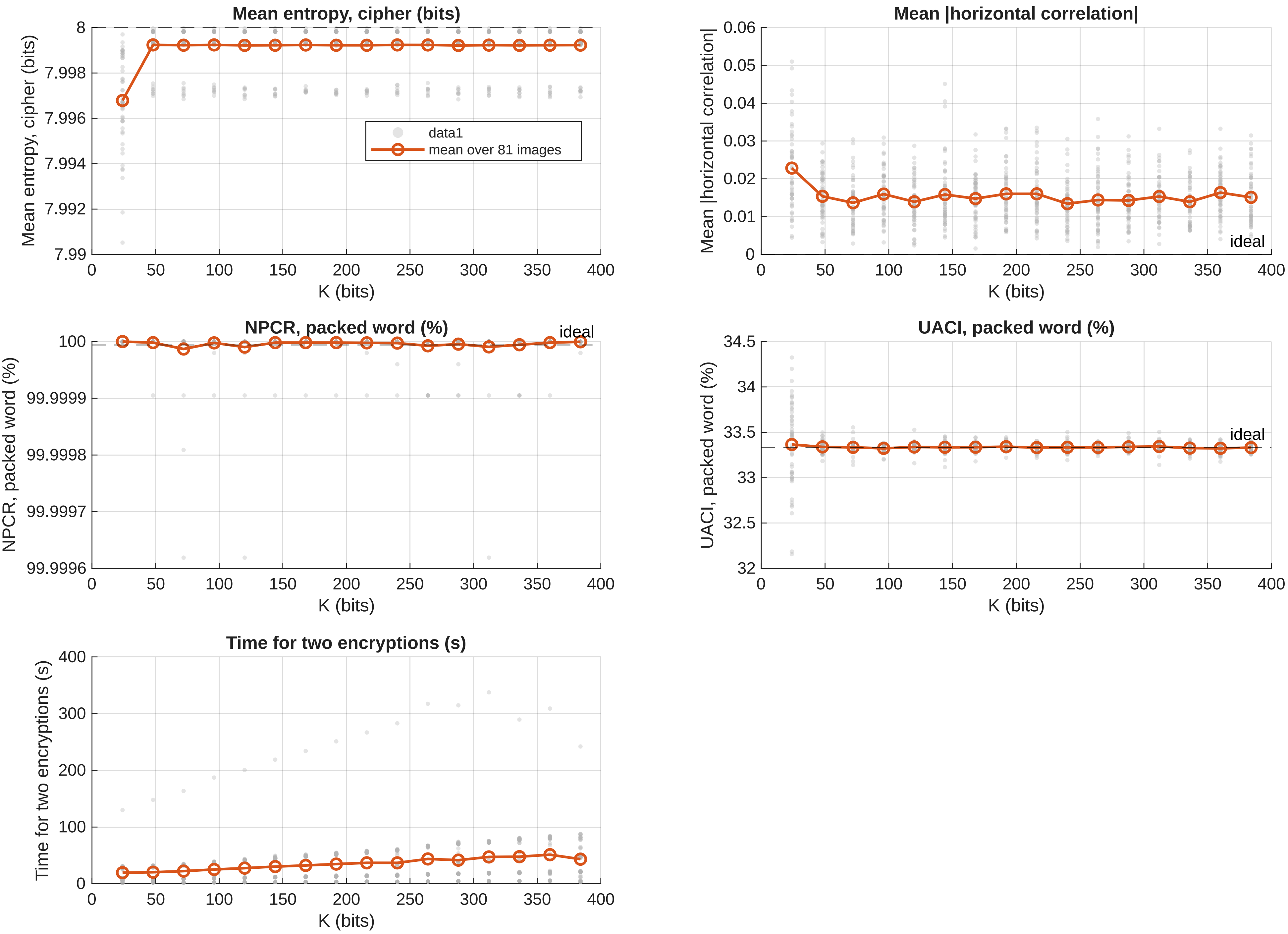
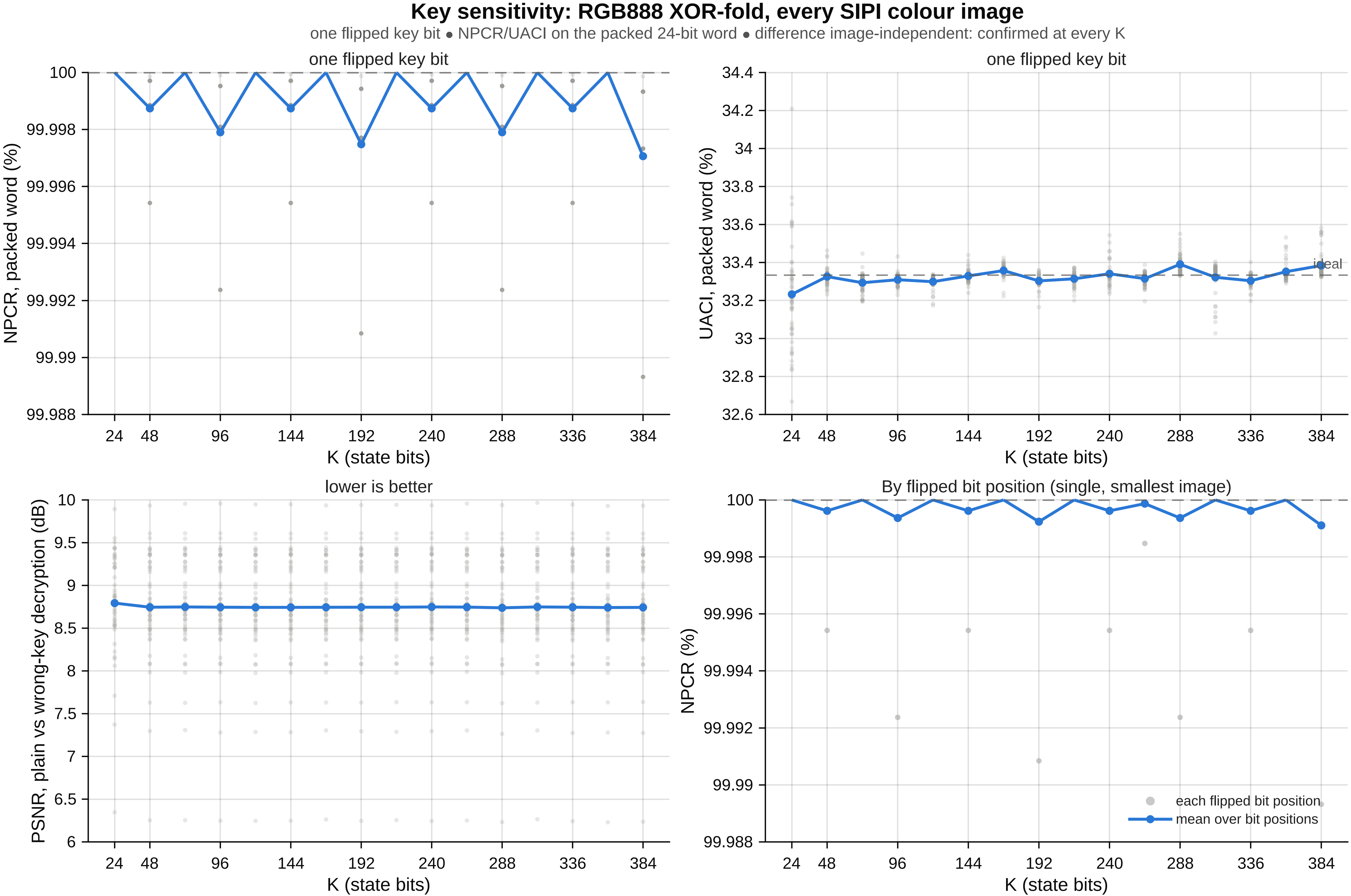
xormap RGB888: one K-bit state XOR-folded into one pixel, every SIPI color image (81), K=24:24:384

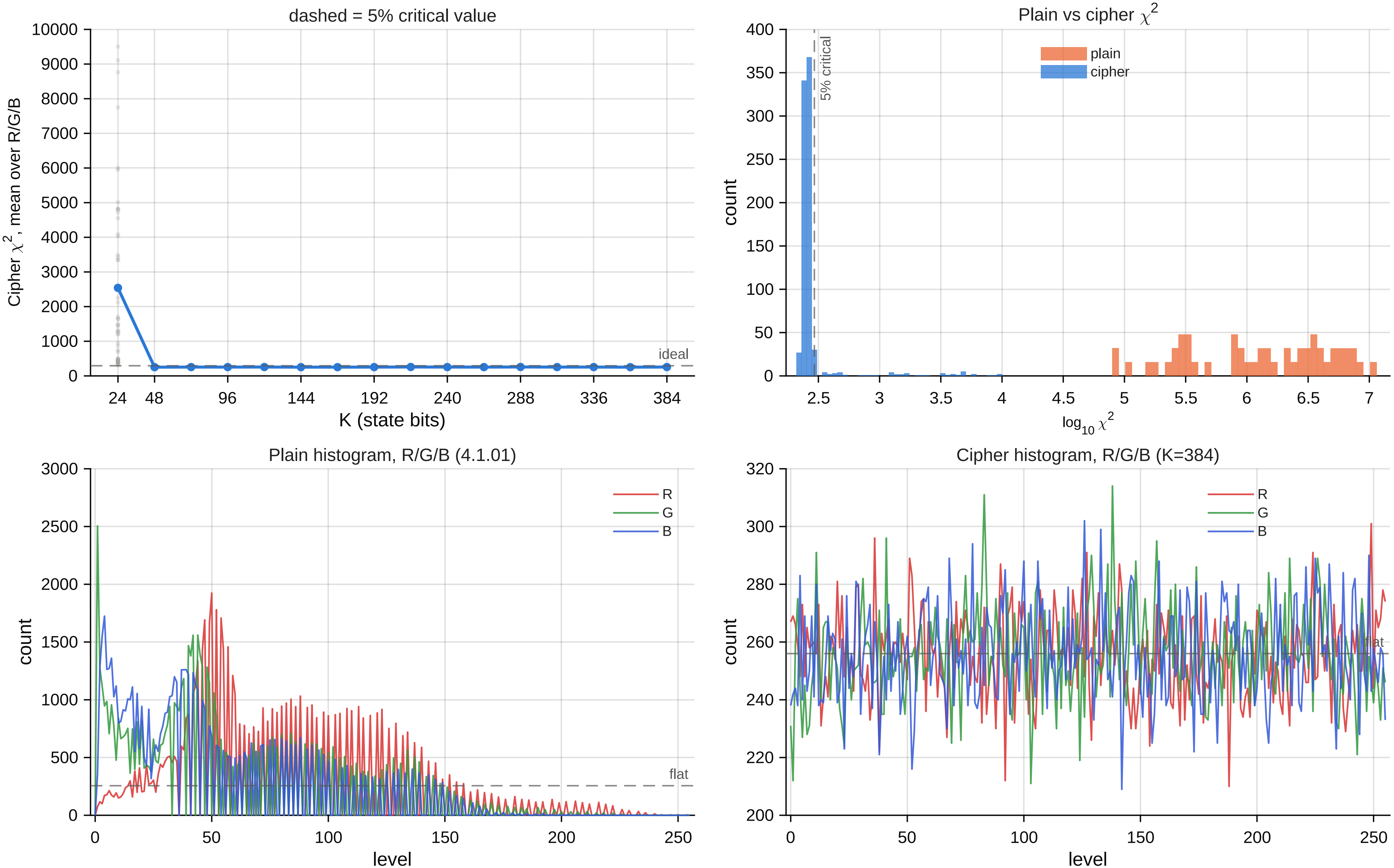


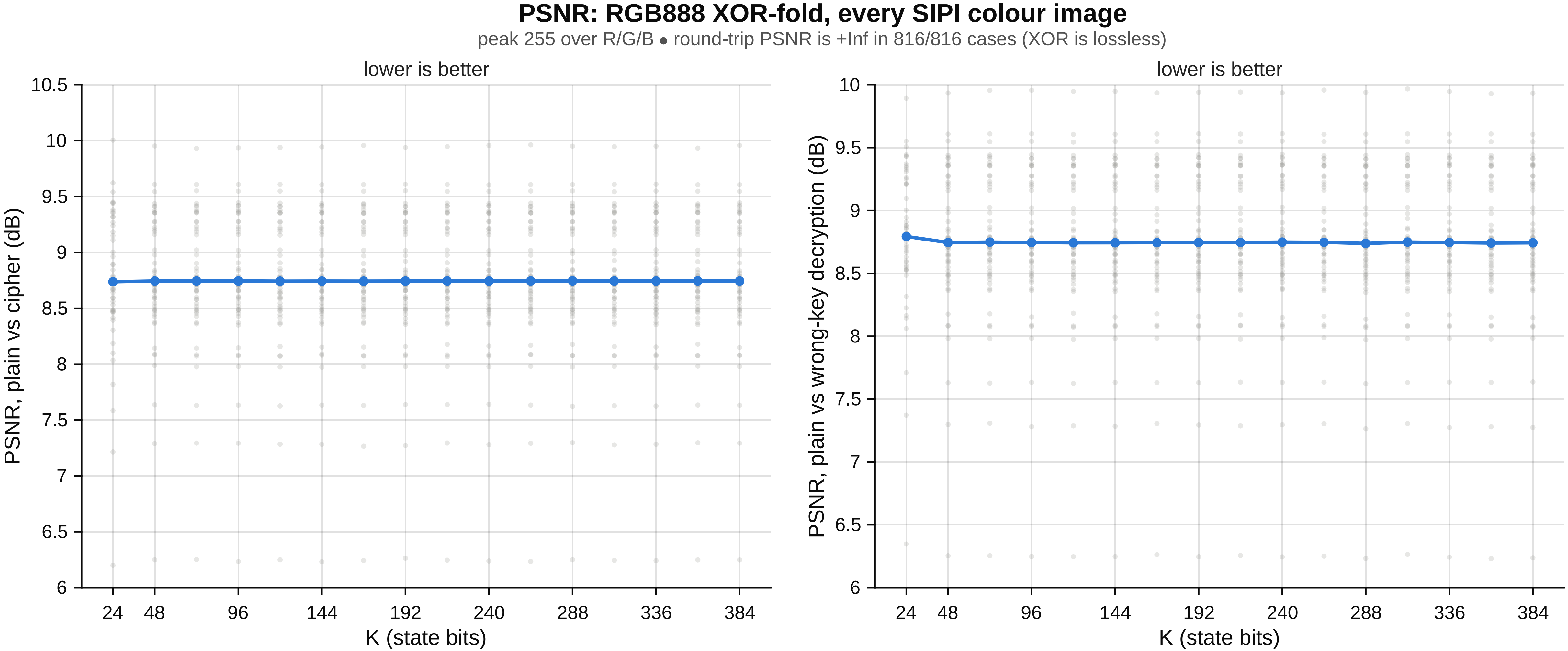


Key sensitivity: RGB888 XOR-fold, every SIPI colour image
one flipped key bit • NPCR/UACI on the packed 24-bit word • difference image-independent: confirmed at every K
one flipped key bit
NPCR, packed word (%)
K (state bits)
one flipped key bit
UACI, packed word (%)
ideal
K (state bits)
lower is better
PSNR, plain vs wrong-key decryption (dB)
K (state bits)
By flipped bit position (single, smallest image)
NPCR (%)
each flipped bit position
mean over bit positions
K (state bits)

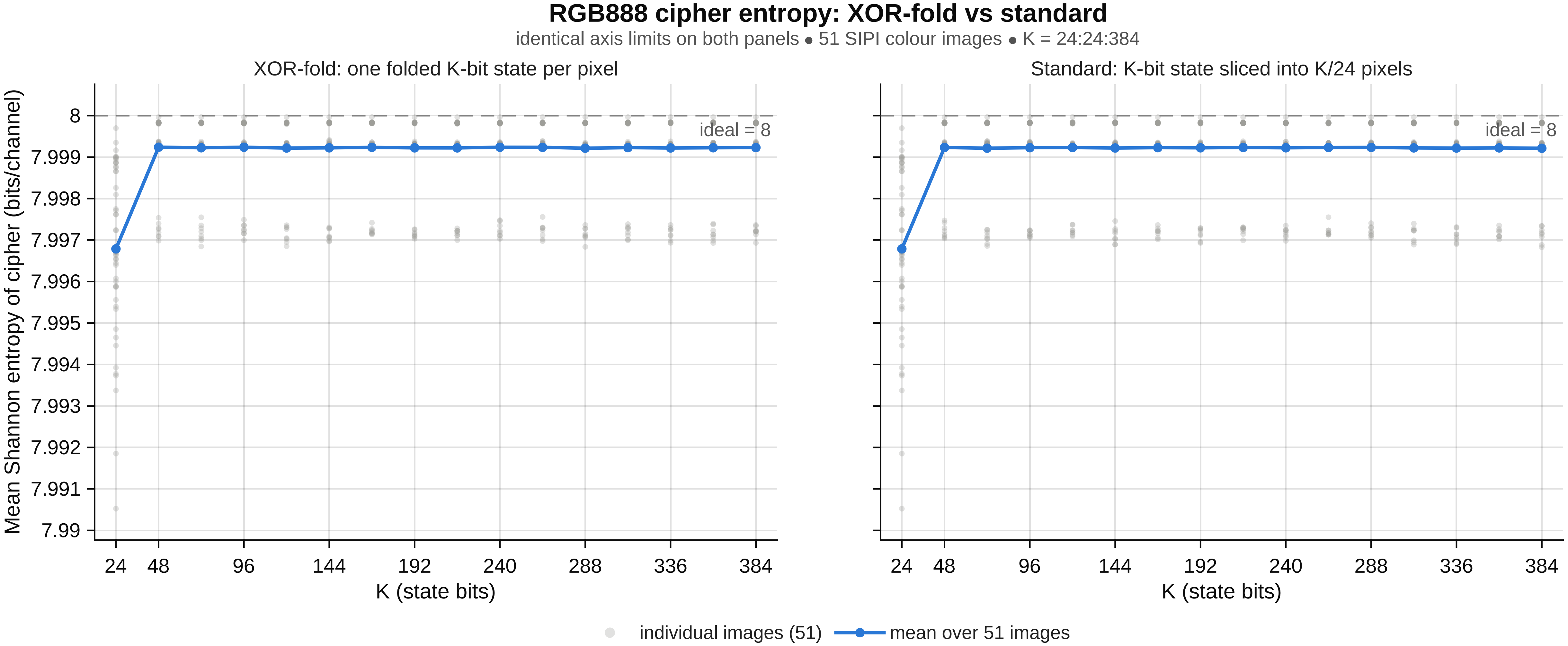
Histogram analysis: RGB888 XOR-fold, every SIPI colour image

χ^2 uniformity per 8-bit channel • cipher passes at α=0.05 in 93.8% of (image, K) cases

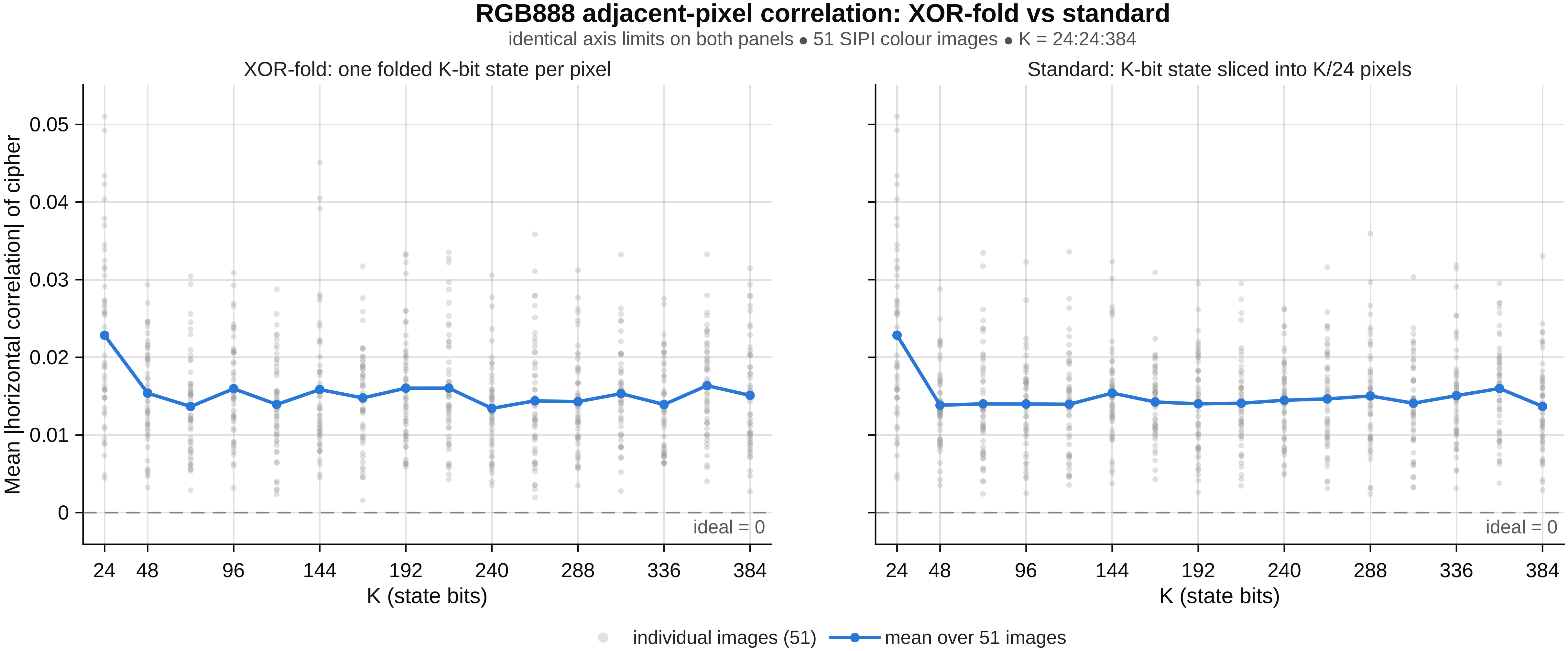




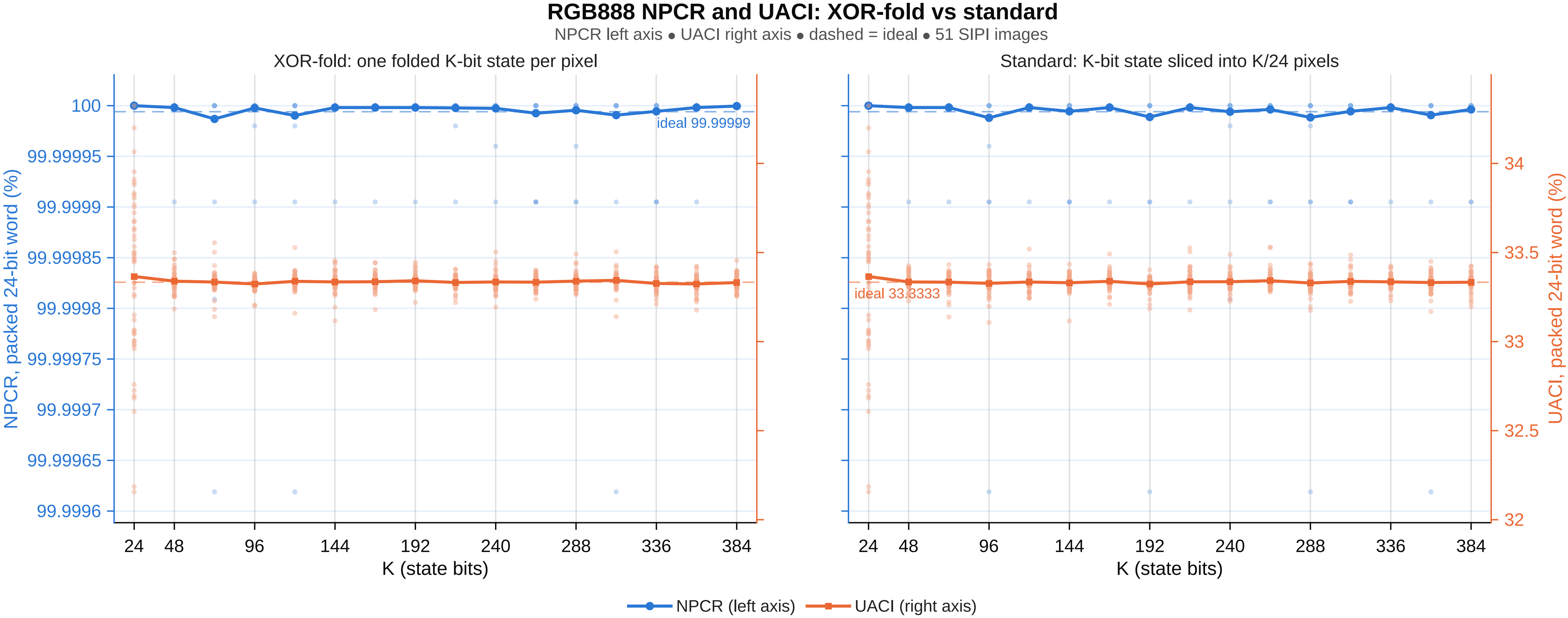
PSNR: RGB888 XOR-fold, every SIPI colour image
peak 255 over R/G/B • round-trip PSNR is +Inf in 816/816 cases (XOR is lossless)
lower is better
PSNR, plain vs cipher (dB)
K (state bits)
lower is better
PSNR, plain vs wrong-key decryption (dB)
K (state bits)
10.5
10
9.5
9
8.5
8
7.5
7
6.5
6
24
48
96
144
192
240
288
336
384



RGB888 cipher entropy: XOR-fold vs standard
identical axis limits on both panels • 51 SIPI colour images • K = 24:24:384
XOR-fold: one folded K-bit state per pixel
Standard: K-bit state sliced into K/24 pixels
Mean Shannon entropy of cipher (bits/channel)
ideal = 8
8
7.999
7.998
7.997
7.996
7.995
7.994
7.993
7.992
7.991
7.99
24
48
96
144
192
240
288
336
384
K (state bits)
individual images (51)
mean over 51 images



RGB888 adjacent-pixel correlation: XOR-fold vs standard
identical axis limits on both panels • 51 SIPI colour images • K = 24:24:384
XOR-fold: one folded K-bit state per pixel
Standard: K-bit state sliced into K/24 pixels
Mean |horizontal correlation| of cipher
0.05
0.04
0.03
0.02
0.01
0
ideal = 0
24
48
96
144
192
240
288
336
384
K (state bits)
individual images (51)
mean over 51 images



RGB888 NPCR and UACI: XOR-fold vs standard
NPCR left axis • UACI right axis • dashed = ideal • 51 SIPI images
XOR-fold: one folded K-bit state per pixel
Standard: K-bit state sliced into K/24 pixels
NPCR, packed 24-bit word (%)
UACI, packed 24-bit word (%)
100
99.99995
99.9999
99.99985
99.9998
99.99975
99.9997
99.99965
99.9996
34
33.5
33
32.5
32
ideal 99.99999
ideal 33.3333
24
48
96
144
192
240
288
336
384
K (state bits)
NPCR (left axis)
UACI (right axis)